Charlotte Grant

Technical Designer · Unreal Engine 5 (Blueprints & C++)

charlottecegrant@gmail.com | https://charlottem7.github.io/Portfolio/ | https://github.com/CharlotteM7 | https://www.linkedin.com/in/charlottecegrant/

Profile

Technical Designer specialising in gameplay systems, puzzle design and level development using Unreal Engine 5, Blueprints and C++.

With a First-Class BSc in Computer Science for Games and an MSc in Advanced Computer Science (Distinction), I combine technical expertise with 14 years of professional experience spanning project management, QA, requirements analysis and team leadership.

Currently working as a Full-Stack Developer, I bring hands-on production coding experience alongside active game development as a volunteer developer on *War of Being*, a narrative-driven adventure game.

Technical Skills

Design
Level design, greyboxing, puzzle design, encounter flow and pacing, player-readable navigation, playtesting and iteration

Unreal Engine 5
Blueprints, C++, Gameplay Framework, Behaviour Trees, Animation Blueprints, UI/UMG, PCG, World Partition

Programming & Tools
C++, C#, Python, JavaScript/TypeScript, Java, PHP. Perforce, Git, Visual Studio, Blender

Games Experience

Volunteer Game Developer | DMTesseracT Ltd **2025 – Present**
War of Being, a sci-fi adventure in current UE5 development: https://store.steampowered.com/app/2475790/

- Design and greybox new level spaces using UE5 blockout workflows
- Program interactable objects, triggers, narrative dialogue trees and scripted Level Sequences in Blueprints
- Shape encounter flow, pacing and player-readable navigation
- Iterate on layouts through internal playtesting
- Collaborate in a small development team using Perforce

Game Projects

Educational Cryptography Puzzle Game (MSc Dissertation)
UE5 • C++ • Blueprints • Systems & Puzzle Design
A third-person puzzle game teaching cryptography concepts through clue-based progression.

- Designed modular puzzle systems (substitution, patterns, decoding) and the clue-based progression that teaches each concept
- Built data-driven puzzle logic for flexible tuning and iteration
- Designed UI feedback, hint systems and accessibility adjustments
- Built quest and interaction systems in Blueprints and C++
- Playtested with 5 players: all with positive results on enjoyment and learning.

Procedural World Generation System (BSc Dissertation)
UE5 • C++ • PCG Tools • Terrain/LOD Systems
A runtime procedural generation system for large, varied terrain, holding 30 fps during runtime mesh generation using LOD and async updates.

- Implemented Perlin-noise terrain generation with biome rules
- Added asynchronous chunk streaming with LOD optimisation
- Generated foliage dynamically from terrain metadata

Multiplayer Pong

SDL2 • C++ • Java TCP Server
A twist on classic Pong with multiplayer networking and expanded gameplay.

- Implemented client/server networking with Java sockets and client-side prediction
- Designed power-ups, particles and extended game modes, with full UI menus

Experience

Full-stack Developer | Liquid Light **Jul 2026 – Present**

- Built the frontend for a gamified campaign site: a badge and progression system that tracked each user's completed steps, with reward states and feedback
- Rebuilt pages and site sections for client websites, turning designs into responsive, accessible interfaces
- Ship production code in PHP, TYPO3, TypoScript and SCSS within an agency team, using GitLab and code review

Freelance Web Developer **Oct 2025 – Present**

- Designed, built and launched websites for independent clients using WordPress and Shopify
- Managed each project end to end, from client requirements to deployment and handover

Earlier Career | Santander Consumer Finance UK & PSA Finance **2011 – 2025**

- Training & Projects Manager (PSA Finance): QA testing, requirements gathering and CRM rollout on digital projects
- Call Centre Manager and Team Leader: led teams in high-volume settings, using data to improve processes
- Technical and data roles (Santander): database investigations, regulatory data work and resolving complex cases on tight deadlines

Education

MSc Advanced Computer Science – Distinction 2024 – 2025
University of Sussex

BSc Computer Science for Games – First Class Honours 2021 – 2024
University of Brighton

Other Qualifications:
Diploma of Higher Education – Business Management (Open University, 2016 – 2018)
PRINCE2 Project Management Foundation & Practitioner (2014)